

Tutorial 1

ECON 101 – Macroeconomics

Fall 2025

Q1. Market Equilibrium and Price Controls

Demand and supply for coffee are:

$$Q_d = 100 - 2P, \quad Q_s = 20 + 3P.$$

- (a) Find the equilibrium price and quantity.
- (b) With a price floor at $P = 20$, compute the surplus.
- (c) With a price ceiling at $P = 10$, compute the shortage.

Solution.

- (a) Equilibrium: set $Q_d = Q_s$.

$$100 - 2P = 20 + 3P \Leftrightarrow 80 = 5P \Leftrightarrow P^* = 16, \quad Q^* = 100 - 2(16) = 68.$$

- (b) At $P = 20$: $Q_d = 100 - 40 = 60$, $Q_s = 20 + 60 = 80$. Surplus = $Q_s - Q_d = 20$ units.
- (c) At $P = 10$: $Q_d = 100 - 20 = 80$, $Q_s = 20 + 30 = 50$. Shortage = $Q_d - Q_s = 30$ units.

Q2. Shifts in Demand from Income Changes

Demand: $Q_d = 500 - 5P + 0.5Y$. Supply: $Q_s = 100 + 3P$.

- (a) If $Y = 200$, find the equilibrium price and quantity.
- (b) If $Y = 300$, find the new equilibrium price and quantity.
- (c) Interpret the income effect.

Solution.

- (a) With $Y = 200$: $Q_d = 600 - 5P$.

$$600 - 5P = 100 + 3P \Leftrightarrow 500 = 8P \Leftrightarrow P^* = 62.5, \quad Q^* = 100 + 3(62.5) = 287.5.$$

- (b) With $Y = 300$: $Q_d = 650 - 5P$.

$$650 - 5P = 100 + 3P \Leftrightarrow 550 = 8P \Leftrightarrow P^* = 68.75, \quad Q^* = 100 + 3(68.75) = 306.25.$$

- (c) Higher income shifts demand right. Both equilibrium price and quantity rise, indicating laptops behave as a normal good.

Q3. Elasticity at a Point

For $Q_d = 200 - 4P$,

- (a) Compute the price elasticity of demand at $P = 30$.
- (b) Classify demand at this price.
- (c) Predict the effect on total revenue from a small price increase.

Solution.

- (a) At $P = 30$, $Q = 200 - 4(30) = 80$. For linear demand, $\varepsilon = \frac{dQ}{dP} \cdot \frac{P}{Q} = -4 \cdot \frac{30}{80} = -1.5$.
- (b) $|\varepsilon| = 1.5 > 1$, so demand is elastic.
- (c) With elastic demand, a price increase lowers total revenue.

Q4. Supply Shift from Input Cost Increase (Graphical and Numerical)

Initial demand and supply:

$$Q_d = 80 - 2P, \quad Q_s = 10 + 2P.$$

- (a) Find and graph the initial equilibrium.
- (b) An input cost increase shifts supply to $Q_s = -10 + 2P$. Find and graph the new equilibrium.
- (c) Explain the comparative statics.

Solution.

- (a) Baseline equilibrium:

$$80 - 2P = 10 + 2P \Leftrightarrow 70 = 4P \Leftrightarrow P^* = 17.5, \quad Q^* = 80 - 2(17.5) = 45.$$

- (b) After the shift:

$$80 - 2P = -10 + 2P \Leftrightarrow 90 = 4P \Leftrightarrow P' = 22.5, \quad Q' = 80 - 2(22.5) = 35.$$

- (c) Higher input costs shift supply left. Equilibrium price rises and equilibrium quantity falls.

Q5. Rent Control: Shortage Quantified (Graphical and Numerical)

Rental market:

$$Q_d = 120 - 2P, \quad Q_s = 20 + 2P.$$

- (a) Find and graph the competitive equilibrium.
- (b) With a rent ceiling at $P = 20$, show the shortage and compute its size.

Solution.

- (a) Competitive equilibrium:

$$120 - 2P = 20 + 2P \Leftrightarrow 100 = 4P \Leftrightarrow P^* = 25, \quad Q^* = 120 - 2(25) = 70.$$

- (b) At $P = 20$: $Q_d = 120 - 40 = 80$, $Q_s = 20 + 40 = 60$. Shortage $= 80 - 60 = 20$ units. On the graph, plot the horizontal gap between Q_d and Q_s at $P = 20$.

Q6. Comparative Statics of a Demand Shift (Graphical and Numerical)

Wheat market:

$$Q_d = 600 - 10P, \quad Q_s = 100 + 5P.$$

- (a) Compute the initial equilibrium.
(b) Demand shifts right to $Q_d = 700 - 10P$. Compute the new equilibrium.
(c) Explain the movements in price and quantity.

Solution.

- (a) Initial:

$$600 - 10P = 100 + 5P \Leftrightarrow 500 = 15P \Leftrightarrow P^* = \frac{500}{15} \approx 33.33, \quad Q^* = 600 - 10(33.33) \approx 266.67.$$

- (b) After shift:

$$700 - 10P = 100 + 5P \Leftrightarrow 600 = 15P \Leftrightarrow P' = 40, \quad Q' = 700 - 10(40) = 300.$$

- (c) A rightward demand shift raises both equilibrium price and quantity.

Notes for graphing across questions: Always mark intercepts. For a linear inverse demand $P = a - bQ$, price intercept $= a$ and quantity intercept $= a/b$. Analogous logic applies for supply.